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AI-augmented construction cost estimation: an ensemble Natural Language Processing (NLP) model to align quantity take-offs with cost indexes

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ABSTRACT

Accurate construction cost estimation is crucial for the financial success of construction projects. Effective financial management in construction heavily relies on a robust and reliable estimation system. Building Information Modeling (BIM) has emerged as a powerful tool, providing precise quantities for various building elements through Quantity Take-Offs (QTOs). Traditionally, Quantity Surveyors (Qs) match these QTOs with cost indexes, a task that is both labor-intensive and prone to errors due to subjectivity and inconsistencies in classification systems and software structures. This paper presents an ensemble Natural Language Processing (NLP)-based method designed to automatically align QTOs with corresponding cost indexes across different building classifications and works. Our proposed method leverages advanced NLP techniques to analyze and align textual descriptions in QTOs with corresponding items in cost indexes. The system was rigorously tested on a high-rise residential building project, where it demonstrated an 82.96% agreement rate with QS-based estimations, highlighting its semantic alignment accuracy. The model also achieved only -2.05% total cost deviation from QS estimates. The proposed Artificial Intelligence (AI)-driven system complements Qs by serving as a notification tool, highlighting discrepancies between human and system-generated estimates. This approach aids Qs in refining their cost assessments, enhancing accuracy and mitigating potential financial risks.

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Construction cost estimation; quantity take-off (QTO); natural language processing (NLP); artificial intelligence (AI); ensemble learning; building information modeling (BIM)

Introduction

Accurate cost estimation is crucial for the success of construction projects, ensuring that projects stay on budget and meet their set objectives within a specific period (Saeidlou and Ghadiminia 2024). Although estimation accuracy can differ at various stages of the design and construction procedures, depending on the available information, the estimation process remains consistently applied across these phases. In the Architecture, Engineering, and Construction (AEC) industry, cost estimation involves a detailed process of predicting the expenses associated with a construction project based on the provided building specifications (AASHTO 2013; Holm and Schaufelberger 2021). This intricate process comprises several crucial steps. The first step is the work breakdown, which involves decomposing a building project into its individual elements (Lee et al. 2014, Holm and Schaufelberger 2021). After the work breakdown, the next phase is preparing the Quantity Take-Off (QTO). QTO involves accurately measuring various building components and identifying their respective materials and work items, which are vital for construction projects (Khosakitchalart et al. 2019; Banihashemi et al. 2022). Once the QTO is complete, the process moves to pricing the work items. This stage includes calculating the actual costs of materials, labor, tools, equipment and subcontractor services for each task or activity

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involved in the final construction, along with accounting for overhead costs and contingency planning (DePico 2012; Tayefeh Hashemi et al. 2020).

In the AEC industry, to formalize an appropriate work breakdown structure, classification and measurement standards are essential for ensuring consistency, accuracy and efficiency in project planning and control worldwide (Lu et al. 2017). Among key international standards, MasterFormat is a comprehensive list of numbers and titles that identify work results and construction practices (CSI 2018). On the other hand, UniFormat is a method for organizing construction information based on functional elements or parts of a facility characterized by their functions rather than the materials and methods used to create them (CSI and CSC 2010). In addition, OmniClass is another classification system that integrates key elements of both MasterFormat and UniFormat to support building lifecycle and project management (CSI 2019). The International Construction Measurement Standards (ICMS) also provides a worldwide framework for classifying, defining, measuring, analyzing and presenting construction costs (ICMS 2016).

In Australia, the construction industry relies on several key standards for classification and measurement. The Australian Standard Method of Measurement of Building Works (ASMM), developed by the Australian Institute of Quantity Surveyors (AIQS), offers detailed guidelines for measuring building works, ensuring consistency in preparing Bills of Quantities (BoQ) (AIQS 2016). NATSPEC, the National Building Specification, also provides an extensive set of specifications for building construction, covering various trades and disciplines, which is utilized for drafting detailed project specifications and ensuring compliance with Australian standards and regulations (NATSPEC 2024). Furthermore, industry-based Construction Cost Indexes (CCIs) such as the AIQS Building Cost Index (BCI) (AIQS 2023), Rawlinsons Australian Construction Handbook (Rawlinsons 2024) and CoreLogic's Cordell Construction Cost Index (CCCI) (CoreLogic 2024) present essential and up-to-date data for monitoring changes and trends in construction costs, supporting precise cost estimation. CCIs serve a dual role in the construction industry. First, they function as indicators of broader market trends, supporting both short- and long-term forecasts of construction cost changes at the industry level. Second, they offer structured cost data that provides a reliable baseline for estimating costs at the project level. This dual function enables CCIs to act both as macroeconomic indicators and as practical reference datasets for cost estimation workflows (Zhang et al. 2019). In this study, the latter function is the primary focus, and CCIs are treated as structured cost reference datasets used to support accurate cost estimation at the project level.

QTO is a process that involves identifying measurement items, gathering dimensional data and computing quantities such as areas and volumes (Liu et al. 2022). Historically, QTOs were performed by Quantity Surveyors (QSSs) using 2D drawings and plans. However, QTO extends far beyond merely reading dimensions; it demands a detailed understanding of construction drawings and specifications to identify all work items, including visible components like walls, footings and slabs, as well as indirect elements such as grading, excavation and formwork that are not always explicitly shown (Popescu et al. 2003). QTO plays a critical role throughout all construction phases. It supports preliminary cost estimation during tendering, helps assess project costs and timelines before construction, aids in planning and forecasting, and ensures financial control during execution. Its accuracy is vital for managing costs and evaluating productivity, directly impacting the project's financial stability (Monteiro and Poças Martins 2013).

Recent efforts have promoted Building Information Modeling (BIM)-based QTO to enhance automation, accuracy and efficiency (Olsen and Taylor 2017; Mattern et al. 2018; Kim et al. 2019), as BIM enables extracting building element quantities through 3D parametric models enriched with lifecycle data (Jafary et al. 2024). However, several limitations still persist in connecting BIM-generated elements to existing classification and measurement standards due to inconsistency between the classification systems used by BIM software and those defined by industry standards. These inconsistencies can lead to discrepancies when matching the identified work items in QTOs with entries in existing CCIs and cost libraries. Moreover, the varying levels of detail and specificity in BIM models can further complicate the alignment process, resulting in potential errors or omissions in cost estimations. This misalignment necessitates manual intervention to ensure accurate cost mapping (Mattern et al. 2018; Akanbi and Zhang 2021; Fazeli et al. 2021).

Regardless of whether QTOs are generated from 2D drawings or through BIM-based 3D workflows, the output consists of free-text work descriptions and quantity values structured under specific classifications. These QTOs are then used as the basis for cost estimation and BoQ, requiring accurate mapping to unit cost data in construction cost databases and CCIs (Steel and Drogemuller 2011; Aram et al. 2014). However, achieving semantic alignment between free-text descriptions and structured cost entries remains a persistent challenge. In real-world practice, Qs frequently face the labor-intensive task of manually cross-referencing descriptions from various sources. This process is time-consuming, subjective, and prone to inconsistencies (Lee et al. 2014; Abanda et al. 2017).

This challenge underscores the pressing need for robust methods that can seamlessly integrate QTOs with standardized cost databases. Addressing this gap, the present study introduces a Natural Language Processing (NLP)-based system that leverages Artificial Intelligence (AI) to automate the alignment of work descriptions in QTOs with corresponding cost items in industry-standard CCIs. While prior studies have explored AI's potential, especially machine learning techniques, in predicting unit costs (Banihashemi et al. 2022) or applying NLP to extract structured data from textual specifications (Akanbi and Zhang 2021), limited research has specifically targeted the alignment of QTOs with CCIs using NLP techniques. Given that both QTOs and CCIs rely heavily on textual descriptions, this system enables accurate semantic matching, offering a practical tool that not only automates but also audits the alignment process. By flagging discrepancies between QS-generated and system-generated matches, it supports Qs in refining cost assessments, improving estimation accuracy and reducing financial risks.

The remaining sections of the paper are structured as follows. First, it reviews previous studies on NLP for cost management and identifies the research gaps to be addressed. Next, it introduces the research method, detailing the preprocessing of the QTO and cost indexes, the development of various NLP-based cost assignment models, and the evaluation strategy. The study then presents the results of applying the developed models to a case study. Finally, the article discusses the findings and concludes with a summary of limitations and future research directions.

NLP for cost management

In the context of Industry 4.0, the construction industry is embracing digitization and AI to enhance automation, productivity and reliability. As data in construction is stored in various electronic text formats, NLP offers intelligent processing of these texts. NLP enables the extraction, representation, retrieval and reasoning of information, making it a crucial technology for advancing intelligence in construction-related fields (Ding et al. 2022; Yang et al. 2024; Ye et al. 2024). Shamshiri et al. (2024) conducted a comprehensive review of text mining and NLP applications in the construction industry. Their findings revealed that the majority of research has concentrated on safety management, automated compliance checking and contract management, while cost management and schedule management have received comparatively less attention.

Among these limited studies, Williams and Gong (2014) explored the use of text mining combined with numerical data to predict construction cost overruns. They developed an ensemble classifier model that processed project descriptions and bid data to predict cost performance, demonstrating the potential of text mining to enhance predictive accuracy. Jafari et al. (2021) developed an automated framework to extract reporting requirements from construction contracts and predict the associated time and costs using NLP and stochastic simulations. The study addressed the challenges of manually extracting requirements from lengthy and complex contractual documents, often leading to underestimating the time and resources needed for reporting. The framework used NLP techniques to automate the extraction of reporting requirements and employed Monte Carlo simulations to predict the overhead costs and durations of report preparation. Tang et al. (2022) proposed an integrated approach to automate the extraction of information from work descriptions for QTO using NLP. The authors developed several named entity recognition models to identify construction activities, materials, building components, product features and measurement units from work descriptions.

In a more relevant study, Akanbi et al. (2019) developed an automated item matching and pricing method to support BIM-based wood construction cost estimation. The method aimed to reduce the manual inputs required in the cost estimation process by matching wood building elements from a

BIM-based design to cost data entries in a cost database. The authors used a combination of Java constructors, HashMaps and NLP techniques to automate the matching and cost extraction processes. The authors then further developed their models in another publication and proposed a semantic NLP-based method to automate the extraction of design information from construction specifications and match the extracted information with items in a cost database to support cost estimation. Their method involved creating a hierarchical semantic model for the building, developing a cost database, and using information extraction and matching algorithms to automate the extraction and matching processes. Although they achieved high accuracy in extraction and matching, their study was limited to a few building works, including structural frames, exterior walls, interior construction, and roofing and waterproofing (Akanbi and Zhang 2021). Expanding upon this line of research, Deza et al. (2022) introduced a machine learning approach to classify construction cost documents into the ICMS. This study focused on leveraging advanced NLP and machine learning techniques to classify BoQ descriptions into standardized ICMS categories, which facilitates global benchmarking and project comparison. By analyzing over 50,000 descriptions from 24 large UK infrastructure projects, the authors demonstrated how advanced machine learning techniques could enhance the standardization and global benchmarking of construction costs.

While the reviewed studies demonstrate the potential of NLP in construction cost-related applications, such as predicting cost overruns from project descriptions (Williams and Gong 2014), extracting work descriptions for QTO preparation (Tang et al. 2022), or classifying BoQ items into international systems like ICMS (Deza et al. 2022), they do not directly address the persistent challenge of semantic alignment between QTO descriptions and standardized CCIs. This issue is especially pronounced during the early design and pre-construction phases, where Qs must manually interpret and match free-text descriptions from QTOs with structured entries in CCIs (Akanbi and Zhang 2021). The task is further complicated by inconsistencies in terminology (Hong et al. 2021), and variations in phrasing among designers (Wohlrab et al. 2019). Poor semantic integration can result in misclassification, scope omissions, and ultimately inefficiencies, rework and financial risk in cost estimation workflows (Xu et al. 2016).

Despite growing interest in NLP for cost-related tasks in the AEC industry, existing approaches are often limited to specific trades or classification schemes, which restricts their generalizability across diverse projects. Aligning QTOs with cost indexes remains a complex and underexplored challenge due to variability in terminology and description styles across regions and design teams. This study contributes to addressing this gap by proposing an ensemble NLP-based model to improve the alignment accuracy between QTOs and CCIs, offering a scalable and more adaptable solution to support reliable cost estimation across varied construction contexts.

Research method

This study focuses on presenting an NLP-based system that automatically assigns costs of different building works from CCIs to work items included in QTOs. CCIs are weighted aggregate measures that track changes in labor, materials and equipment costs over time, providing stakeholders in the construction industry with tools to estimate cost fluctuations and manage economic dynamics across various projects (Cao et al. 2015; Elfahham 2019). In addition, CCIs contain comprehensive lists of work items categorized under various construction classifications, each accompanied by standardized unit descriptions and cost rates, offering a reliable baseline for project-level cost estimation (Zhang et al. 2019). This study focuses on the latter function of CCIs. Specifically, it uses the AIQS BCI, which not only tracks changes in the prices of construction work items at the municipal level and is published quarterly but also provides structured datasets listing classified work items along with unit descriptions and corresponding cost rates (AIQS 2023). Aligning QTO descriptions with such cost datasets is essential for producing reliable and standardized cost estimates. Accordingly, a comprehensive Python-based system is developed to conduct various steps of the alignment process. The system's performance is tested and evaluated using a case study approach. The methodology encompasses several key steps, which are elaborated on below.

Although the preparation of QTOs is beyond the scope of this paper, we first outline the preliminary steps involved in QTO preparation and then focus on the detailed methodology used to develop the NLP-based cost alignment system. The initial step in the cost estimation process involves acquiring all

relevant documentation. This includes 2D plans, 3D BIM models or other relevant documents, which serve as the basis for identifying and quantifying building components. Software tools such as AutoCAD, Revit, ArchiCAD and Navisworks are commonly used to facilitate the preparation and management of these 2D and 3D datasets (Sepasgozar et al. 2022).

The next phase involves importing the building plans into software that supports the measurement and quantification of building components. Tools like Bluebeam Revu, PlanSwift, Autodesk QTO and CostX are instrumental in this process, as they provide detailed measurements and QTO generation capabilities (Babatunde et al. 2020; Wahab and Wang 2022). It is crucial at this stage to adhere to established measurement standards and to use a consistent classification system that will be pivotal for accurate cost matching through relevant cost indexes.

Once the QTO is prepared, the primary contribution of this research begins with matching the work items across different building classifications to the corresponding items in the CCIs. This study focuses on QTOs derived according to ASMM in the CostX and the AIQS BCI as the CCI reference. However, the framework can be adapted for other classification and measurement standards and QTOs provided by other tools, with minor adjustments. The overall workflow of the proposed NLP-based cost alignment framework is illustrated in Figure 1, which connects all key stages of the system from data inputs to model evaluation and result interpretation. The different stages of the system are further explained in the sub-sections below.

Preprocessing of QTO and BCI

The QTO file typically contains extra information, such as unnecessary fields, redundant rows, and some that need to be consolidated, as well as empty rows. It may also include details of different work items for various classifications together. Therefore, before using these files as input for an NLP-based price-matching system, it is necessary to perform preprocessing steps to transform the QTO file into a structured and clean format, facilitating accurate cost alignment in the subsequent phases. To this end, a QTO-preprocessing program was developed in Python to read the QTO spreadsheet prepared in CostX and output cleaned and structured separate spreadsheets for different classes based on the ASMM classifications. The developed program undertakes the following steps to preprocess the QTO data:

- **Data cleaning:** The initial dataset is cleaned by removing unnecessary fields, empty and redundant rows to focus solely on the essential data.
- **Segregation of classifications:** The data is filtered to extract the rows that represent significant classification headings in the QTO. Each classification is isolated into a separate dataset. For each classification, rows are extracted based on their codes and descriptions, ensuring that all relevant data for a specific classification are grouped together.
- **Description enhancement:** The descriptions of work items are refined by consolidating relevant information from adjacent rows, enhancing the clarity and detail of each entry for subsequent NLP-based analysis.

Furthermore, the cost data from the AIQS BCI was processed to ensure it was in a suitable format to match the QTO data. Costs associated with different classifications were extracted from the BCI and

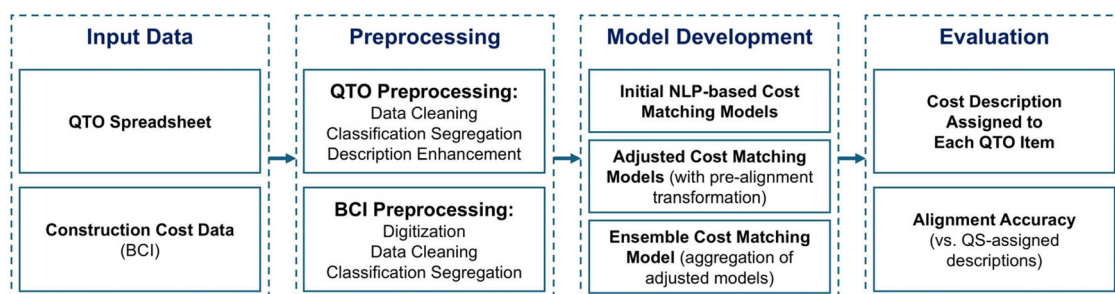


Figure 1. Overview of the developed NLP-based cost alignment framework.

organized into separate spreadsheets. The unit rates in the BCI encompass all the provisions outlined in the ‘Measurement & Prices’ clauses within each section of the ASMM (AIQS 2023).

Cost assignment models

In order to develop an accurate system for matching items from the QTO with corresponding cost entries from the BCI, a series of NLP-based models that analyze textual descriptions and align them with the relevant cost data were established through a Python program.

Initial similarity matching models

According to Wu et al. (2022), when reviewing NLP uses in the construction industry, multi-model, hybrid language processing approaches can play a key role in enhancing text analysis accuracy. Accordingly, four NLP-based models with distinct characteristics and complementary roles were initially employed in this study to match QTO descriptions with BCI cost descriptions, including spaCy, Bidirectional Encoder Representations from Transformers (BERT), Word2Vec and GloVe. These four models were selected based on their complementary strengths, based on the following considerations:

- *Rule-based NLP*: Rule-based methods have demonstrated strong capabilities in various construction applications, including compliance checking (Zhang and El-Gohary Nora 2016), contract risk analysis (Lee et al. 2019), safety management (Xu et al. 2021), and jobsite sensing and monitoring (Ren and Zhang 2021). This study selected spaCy as a representative rule-based NLP model due to its lightweight architecture, efficiency and robust linguistic processing capabilities. SpaCy offers pre-trained language models that capture semantic nuances and support phrase-level pattern matching using token attributes and syntactic dependencies (Altinok 2021, Fantechi et al. 2021). This functionality facilitates the effective measurement of textual similarity between QTO and BCI descriptions. In the developed Python program, the pre-trained model ‘en_core_web_sm’ was used to provide access to comprehensive word embeddings and linguistic annotations.
- *Contextual NLP (transformer-based)*: BERT, a transformer and deep learning-based model, was chosen for its advanced capability in contextual representation, making it well-suited for capturing the nuances of construction-related documents. BERT generates dynamic word embeddings that reflect the surrounding context, which is particularly valuable when interpreting domain-specific terms that may vary across different building tasks or components (Devlin et al. 2018, Peinelt et al. 2020). As Ding et al. (2022) noted in their review of NLP applications in construction, models like BERT offer a high level of text understanding and achieve state-of-the-art performance across NLP tasks. Several pioneering studies have demonstrated BERT’s effectiveness in this domain, such as Zhong et al. (2020), who developed a BERT-based natural language generation model for automated question answering from technical building documents. Our study implemented the BERT-based model using the pre-trained ‘bert-base-uncased’ version, which is widely used for English-language tasks and has demonstrated strong generalization across domains (Geetha and Karthika Renuka 2021).
- *Contextual embedding (static)*: The third model used for matching was Word2Vec, a widely adopted NLP technique based on statistical contextual embeddings that converts words into continuous vector representations. Word2Vec is well known for its ability to analyze large corpora to learn word associations and contexts, capturing semantic relationships between words. This makes it particularly useful for text similarity tasks, including those in the construction industry, such as classifying exchange information requirements (Mitera-Kielbasa and Zima 2024) or construction accident causes (Zhang 2022). In this study, Word2Vec embeddings were loaded from Google’s pre-trained model trained on the Google News dataset (Mikolov et al. 2013).
- *Count-based embedding*: The last model employed was GloVe, a count-based embedding method that derives semantic meaning from the global co-occurrence of words across a corpus (Sakketou and Ampazis 2020). Unlike context-based models such as Word2Vec, GloVe focuses on capturing broader statistical patterns in word usage, making it efficient for representing general semantic associations. This has been demonstrated in studies such as Jeon et al. (2021), which used GloVe to extract construction quality requirements from textual specifications. In the developed system in our

study, GloVe embeddings were loaded via spaCy's 'en_core_web_md' model, which provides access to pre-trained vectors for a medium-sized English vocabulary.

By integrating spaCy's rule-based method, BERT's deep contextual understanding, Word2Vec's context-driven embeddings, and GloVe's count-based representation, this study leveraged a diverse set of NLP paradigms to enhance semantic matching between QTO and BCI descriptions.

Despite their underlying technical differences, all models followed a similar high-level workflow (Figure 2): they began by loading pre-trained language models, tokenizing QTO and CCI descriptions, generating text embeddings, and calculating the cosine similarity between these embeddings to determine how closely related they are in semantic space (Singh and Shashi 2019). The BCI item with the highest similarity score was then selected as the best match for each QTO item.

Adjusted similarity matching models

Following the initial similarity matching models, the study developed adjusted versions of the four models to improve the accuracy and reliability of the cost assignment process. These adjusted models incorporated processing texts before being put into the models and adjusting the similarity scores. Each adjusted model was built upon its corresponding initial model, implementing specific enhancements tailored to address the unique challenges presented by the QTO and BCI data.

For adjusting the four models, spaCy, BERT, Word2Vec and GloVe models, several text adjustment operations were implemented, as presented in Figure 3. These adjustments aimed to normalize the text and reduce noise, ensuring that the descriptions in the QTO and BCI were standardized for more accurate comparisons. The key preprocessing techniques included:

- **Lowercasing:** Converting all text to lowercase, ensuring that the comparisons are case-insensitive, treating words like 'Concrete' and 'concrete' as identical.



Figure 2. General workflow for similarity matching models.

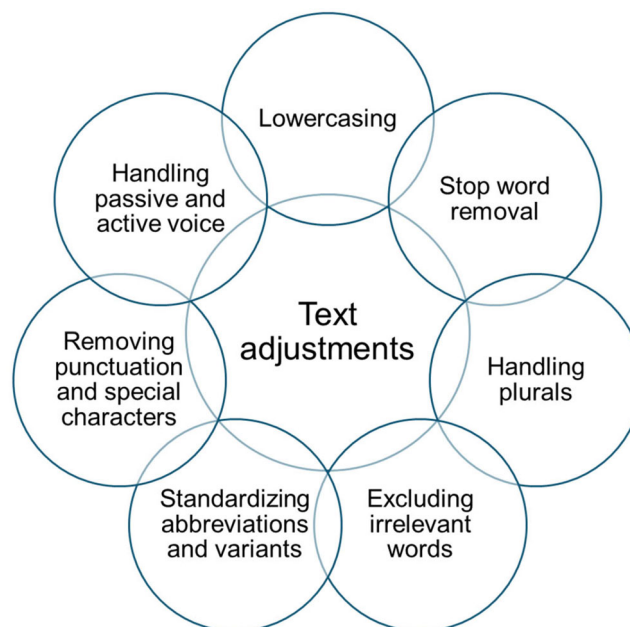


Figure 3. Text processing strategies for adjusted similarity matching models.

- Stop word removal: Stop words, such as ‘the’, ‘is’, ‘in’ and ‘and’ are common words that do not carry significant meaning for the task of matching descriptions, were removed.
- Handling plurals: Plural forms of words were normalized to their singular forms. For instance, ‘windows’ were converted to ‘window.’
- Excluding irrelevant words: Words deemed irrelevant for the context of matching, such as ‘level’ and ‘including,’ were excluded from the text. These words might be common in the descriptions but do not contribute meaningfully to the task of matching work items with cost entries.
- Standardizing abbreviations and variants: Abbreviations and variations in word forms were standardized. For example, ‘m2’ and ‘sq meters’ were converted to a consistent format.
- Removing punctuation and special characters: Punctuation marks and special characters, which do not add value to the semantic meaning of the text, were removed.
- Handling passive and active voice: Descriptions in passive or active voice were normalized to reduce variability. For instance, ‘Concrete was poured’ was standardized to ‘Concrete poured’ to match semantically identical but structurally different descriptions.

Subsequently, an adjustment coefficient was introduced for each of the four adjusted models. This coefficient was designed to refine the similarity scores by considering the frequency and context of key terms, as well as the structure of the descriptions. The general steps for incorporating the coefficient are presented in Figure 4 and are explained as follows:

- Identification of key terms: Key terms in both the QTO and BCI descriptions are identified. These terms, which carry substantial semantic weight, are crucial for the matching process and are identified by analyzing the text and filtering out common words between each work description in QTO and BCI.
- Calculation of key term overlap: The overlap of key terms between the QTO and BCI descriptions is calculated. The number of common key terms in both descriptions and their relative positions are determined. Accordingly, a measure indicating the level of analogy between work descriptions is calculated in terms of their content.
- Calculation of adjusted similarity score: The adjusted similarity score is obtained by multiplying the base similarity score by the overlap coefficient.

Ensemble model

Recent literature highlights the growing importance of ensemble models in NLP, particularly for tasks involving complex semantic matching. Ensemble models have emerged as powerful mechanisms, combining diverse model outputs to better navigate human language’s subtleties and complexities (Jia et al. 2023). These models integrate individual predictive strengths to counteract limitations and consistently improve performance across different tasks in text analysis, such as classification, semantic similarity and summarization (Devi and Sivakumar 2021, Mohammed and Kora 2022). As noted by Mohammed and Kora (2023), ensemble learning is often preferred over single-model approaches because it yields stronger and more reliable predictions by aggregating multiple learning algorithms. Furthermore, experimental results from recent applications, such as multi-domain summarization using hybrid ensemble word embedding models, confirm improved accuracy and overall efficiency compared to standalone methods (Devi and Sivakumar 2021). In light of this evidence, an ensemble model was also developed, leveraging the strengths of multiple adjusted similarity matching models by combining their predictions to achieve a consensus-based cost assignment through the steps presented in Figure 5 and outlined below:

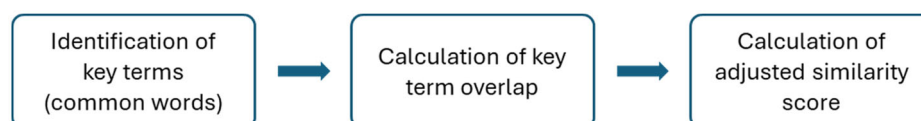


Figure 4. Incorporation of coefficients for adjusting similarity scores.

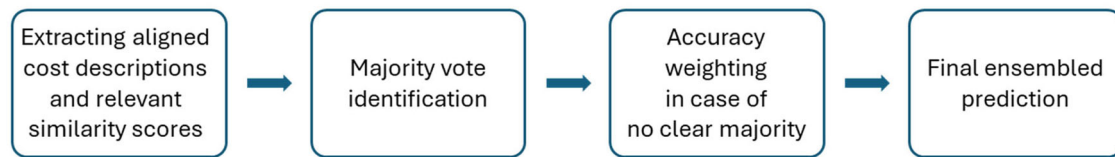


Figure 5. Ensembling process for cost assignment based on adjusted models.

- **Extracting aligned cost descriptions and relevant similarity scores:** Based on the four different adjusted models in the previous stage, predicted BCI cost descriptions and their corresponding similarity scores are extracted for each work description in the QTO.
- **Majority vote identification:** A majority vote mechanism is employed based on counting the frequency of each predicted BCI description across the adjusted models and selecting the one with the highest frequency as the final match of the ensemble model.
- **Accuracy weighting:** In cases with no clear majority, the similarity scores are weighted based on each model's agreement with the QS's alignments for the classification under consideration. The model with the highest agreement is given more weight in the final decision.
- **Final ensemble prediction:** The BCI description with the highest majority vote or weighted score is selected as the final match for each QTO description.

Evaluation

To validate the performance of each model, the percentage of agreement between the cost descriptions assigned by a QS and the predictions made by the different models is calculated for each classification. This evaluation process is conducted for all nine models, including the four initial models (spaCy, BERT, Word2Vec, GloVe), the four adjusted models, and the ensemble model. The agreement percentages are determined by comparing the model-assigned cost descriptions with those assigned by the QS for various QTO work descriptions across different building classifications in the ASMM.

Case study results

The case study presented in this section focuses on the construction project for a 13-level building, encompassing a comprehensive development that includes 128 residential apartments, a retail tenancy, parking facilities for cars and bicycles, and a podium garden. The project's scope measurements are in accordance with the ASMM in the CostX, specifically adhering to the guidelines set out in the Fifth Edition from January 1990. The site for the project covers an area of 3371.36 square meters, accommodating the extensive facilities required for the residential, commercial and recreational spaces planned within the building. This case study illustrates the application of the developed NLP-based cost assignment models across various building classifications as outlined in the ASMM. In the following, we will delve into the results and analysis, highlighting the performance and effectiveness of each model in aligning QTO descriptions with the corresponding cost items from the BCI. By examining this real-world project, the study provides insights into the practical implications of the proposed cost assignment system.

For the case study, the QTO spreadsheet required several preprocessing steps to prepare it for integration with the price matching system. This involved cleaning and organizing the data to ensure it was structured appropriately for accurate cost matching with the BCI. Initially, the QTO contained a total of 34 classifications, ranging from General Notes to External Works. However, the developed Python-based preprocessing program identified nine classifications containing three or fewer quantified work items. These classifications included General Notes, Preliminaries, Overheads and Margin, Demolition, Woodwork, Electrical Services, Mechanical Services, Fire Protection Services, and Vertical Transportation Services. Due to the limited quantified work items, these classifications were excluded to avoid introducing uncertainty and variability into the model's performance evaluation.

Additionally, it was noted that no direct cost descriptions were available in the BCI for five classifications of Post Tensioning, Partitions and Wall Linings, Joinery, Whitegoods and Appliances and

Special Equipment, leading to their exclusion from the system's input. The classifications of Carpet Floor Finish and Floor and Wall Tiling in the QTO were also merged for consistency and matched with the items under the Finishes classification in the BCI. Consequently, the preprocessing process yielded 19 cleaned and organized classification spreadsheets from the initial QTO, which were then utilized as inputs for the cost matching model.

On the BCI side, the relevant document from December 2023 was digitized, and various spreadsheets corresponding to the classifications within the BCI were created. The BCI annually provides a comprehensive dataset that includes fields such as Classification, Unit, and the prices for different building work items in various capital cities across Australia, namely Sydney, Brisbane, Melbourne, Perth, Adelaide and Canberra. For this study, the prices associated with Melbourne were specifically considered to align with the case study's geographical context.

After running the developed program across 19 classifications, nine distinct NLP-based similarity matching models, ranging from initial models to adjusted ones, and culminating in the ensemble model, were employed to predict a cost description and the corresponding cost from the BCI for each item in the QTO. Table 1 lists the 19 considered classifications and the number of work items processed by the models per each classification. In total, 592 items were successfully assigned a cost description and a corresponding price through the developed NLP-based system.

The alignment percentage of each model for each classification was computed by comparing the models' predictions with the cost descriptions assigned by the QS based on the BCI. Figure 6 presents a heatmap that visualizes the alignment percentages achieved by each model for the different classifications. The color gradient in the heatmap transitions from red, indicating a 0% alignment, to green, indicating a 100% alignment. This visual representation effectively highlights the enhancement in alignment accuracy, progressing from the initial models through the adjusted models, and ultimately to the ensemble model. The heatmap underscores the improvement in model performance, with the ensemble model achieving the highest alignment percentages across most classifications, thereby demonstrating its robustness and effectiveness.

Additionally, Figure 7 illustrates the cost alignment percentage across different classifications based on the ensemble model. This figure provides a clear visual representation of how effectively the ensemble model assigned cost descriptions to various QTO items. The line chart reveals that the ensemble model achieved a 100% alignment for the Groundworks, Reinforcement and Stonework classifications. This indicates that all QTO items within these classifications were matched with cost descriptions from the BCI that exactly corresponded with those assigned by the QS. The ensemble model also demonstrated strong performance across other classifications. Eleven classifications achieved an alignment percentage exceeding 80%, showcasing the model's high accuracy and reliability in cost description

Table 1. Number of considered items per classification in the NLP-base cost alignment models.

Classification	Number of considered items in the models
Groundworks	12
Piling, Ground Anchor and Infill Wall	5
Concrete	108
Formwork	78
Reinforcement	80
Waterproofing and Tanking	13
Masonry	5
Stonework	16
Structural Steel	6
Metalwork	12
Glazing	12
Roofing	5
Suspended Ceilings	10
Facade	153
Doors	12
Finishes	12
Painting	12
Hydraulics	17
External Works	24
Total	592



Figure 6. Heatmap showing the alignment percentages of cost descriptions based on different NLP-based models with QS's judgments across various classifications.

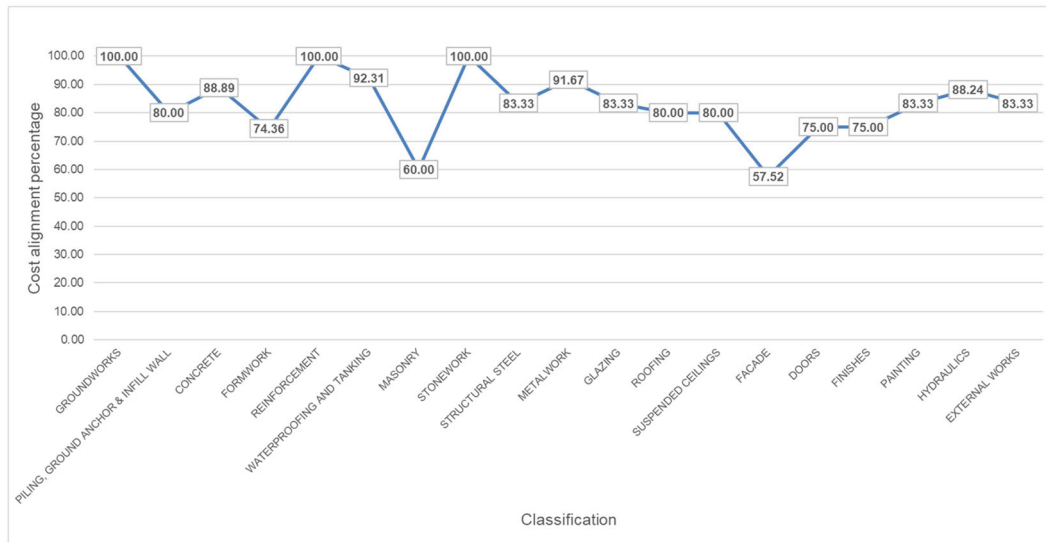


Figure 7. Cost alignment percentage of different classifications based on the ensemble model.

assignment, and three classifications had alignment percentages between 70% and 80%, indicating a solid performance of the ensemble model in these areas as well. However, the model's performance in the Masonry and Facade classifications was notably lower, with alignment percentages around 60%. These results highlight areas where the model struggled to match QTO descriptions with corresponding BCI cost entries accurately. The reasons for these discrepancies will be explored in detail in the discussion section of the paper to identify potential improvements and refinements for the models.

Overall, the ensemble model demonstrated a high level of conformity in aligning cost descriptions, with 82.96% of the 592 items receiving similar cost descriptions when compared to those assigned by

Table 2. Cost comparison between ensemble model and QS estimations across various classifications.

Classification	Ensemble model cost estimation	QS cost estimation	Price difference	Price difference percentage (%)
Groundworks	\$199,441	\$199,441	\$0	0.00
Piling, Ground Anchor and Infill Wall	\$771,414	\$762,414	\$9000	1.18
Concrete	\$3,333,739	\$3,316,011	\$17,728	0.53
Formwork	\$5,296,053	\$5,537,214	−\$241,161	−4.36
Reinforcement	\$2,140,180	\$2,140,180	0	0.00
Waterproofing and Tanking	\$658,005	\$667,437	−\$9432	−1.41
Masonry	\$269,580	\$197,855	\$71,725	36.25
Stonework	\$112,079	\$112,079	\$0	0.00
Structural Steel	\$762,491	\$766,478	−\$3987	−0.52
Metalwork	\$1,174,742	\$1,189,642	−\$14,900	−1.25
Glazing	\$877,780	\$890,990	−\$13,210	−1.48
Roofing	\$48,543	\$48,828	−\$285	−0.58
Suspended Ceilings	\$930,570	\$957,360	−\$26,790	−2.80
Facade	\$13,017,357	\$13,282,835	−\$265,478	−2.00
Doors	\$1,850,110	\$2,027,665	−\$177,555	−8.76
Finishes	\$1,490,180	\$1,528,241	−\$38,061	−2.49
Painting	\$1,154,181	\$1,162,167	−\$7,986	−0.69
Hydraulics	\$4,702,721	\$4,805,311	−\$102,590	−2.13
External Works	\$1,129,290	\$1,160,767	−\$31,478	−2.71
Total	\$39,918,455	\$40,752,915	−\$834,460	−2.05

Table 3. Example matches of QTO item descriptions with assigned descriptions by the ensemble model.

Item description in QTO	Description assigned by the ensemble model based on BCI descriptions
'DETAILED EXCAVATION Excavation including allowance for working space and/or maintaining faces of excavation Pile caps not exceeding 1m total depth'	'EXCAVATION Trench excavation Not exceeding 2m deep'
'INTERNAL BLOCKWORK WALL Internal concrete blockwork including wall stiffeners as required, including forming door opening and other associated works 100mm thk'	'BLOCKWORK Standard 400 x 200mm solid grey concrete blocks in: 100mm wall'
'STRUCTURAL STEEL SUPPORT Allow structural steel support with paint finish to screens and blockwall Facade FA:05 carpark louvre screen (40kg/m ²)'	'STRUCTURAL STEEL Universal beams and columns with welded fitments for simple connections: Up to 60kg/m'
'INTERNAL PLASTERBOARD WALLS Acrylic paint to plasterboard wall lining at external wall façade'	'PLASTERED AND SIMILAR WALL AND CEILING SURFACES Seal and two coats acrylic paint'
'SUSPENDED PLASTERBOARD CEILINGS Suspended flushed plasterboard ceiling to Ground Floor Lobby, Lift Lobby, Entry Air Lock, mail area and Office'	'SUSPENDED CEILINGS Linings: Flush plasterboard ceiling'

the QS. This indicates that the model was able to closely match the expert judgments for a significant majority of the items, validating the effectiveness of the NLP-based approach in automating cost assignment processes.

Table 2 provides a comparison of the cost estimations derived from the ensemble model against the cost estimations made by the QS for each classification. The costs are represented in Australian Dollars (AUD), and the table also highlights the absolute and percentage differences between the two sets of estimations for each classification. The results demonstrate the high level of conformity in cost estimation achieved by the ensemble model, with a total price difference percentage of just −2.05% across all 19 classifications. An intriguing observation is that despite the low alignment accuracy for the Facade classification, the cost estimation by the ensemble model only differed by 2% from the QS's estimation.

Following the evaluation of the models, some examples are presented in Table 3 to illustrate how the ensemble model's cost descriptions based on the BCI descriptions align with the work descriptions in the QTO. These examples showcase the NLP-based systems's ability to effectively translate detailed item descriptions into standardized cost descriptions.

Discussions and conclusions

This study proposed an NLP-based framework for analyzing and aligning textual descriptions in QTOs with standardized CCIs, with a particular focus on AIQS BCI in Australia. The method was implemented and tested on a real-world case involving 592 QTO entries across 19 distinct ASMM-based classifications from a 13-level residential construction project. The goal was to improve semantic

alignment between QTO work descriptions and cost database entries as an error-prone and time-consuming task traditionally handled manually by Qs and often marked by inconsistency, subjectivity and inefficiency (Lee et al. 2014; Abanda et al. 2017).

To explore this, nine NLP models were developed and compared, including four baseline models (spaCy, BERT, Word2Vec and GloVe), four adjusted variants, and a final ensemble model. Model evaluation was conducted by calculating agreement percentages between NLP-predicted cost matches and those assigned by a professional QS in the known BoQ. The ensemble model demonstrated a significant improvement in performance, achieving an overall alignment accuracy of 82.96%, with most classifications exceeding 80%.

While this implies about 17% disagreement rate, the study also found that the overall price difference between the NLP-generated BoQ and the QS-assigned BoQ was only -2.05% across all classifications, indicating a strong level of cost conformity despite occasional mismatches in specific item pairings. It is important to recognize that cost estimation is inherently subject to variability, and perfect alignment, even among expert Qs, is rarely achieved due to differences in interpretation, experience and skillset. Although direct comparison with other studies is difficult given the uniqueness of our NLP-based alignment approach, reference to related AI-based cost estimation research helps contextualize our results. For example, Swei et al. (2017) reported R^2 values ranging from 0.12 to 0.78 across different pavement types in parametric cost estimation models, with notable heteroscedasticity. Fazeli et al. (2021) showed deviation percentages of 5% to 31.5% across different building elements when comparing BIM-based and traditional cost estimates against actual costs. Similarly, Banihashemi et al. (2022) found variation rates of 13%–32% across elements when comparing their machine learning based estimation model using Adaptive Neuro Fuzzy Inference System (ANFIS) and BIM-based baseline estimates. Against this backdrop, our ensemble model's 82.96% alignment rate and only -2.05% deviation in total price estimation compare favorably and demonstrate both reliability and practical viability. These results support the claim that our system provides a meaningful and accurate enhancement to current cost estimation practices, especially given its automation of a task that is otherwise time-consuming and prone to subjectivity.

The results also highlight the potential of combining multiple NLP paradigms for capturing semantic nuances in construction-related textual data. Accordingly, the study validates the value of ensemble approaches in NLP applications in the construction sector. By integrating the strengths of four foundational NLP techniques, spaCy for phrase pattern matching, BERT for contextual embeddings, Word2Vec for word associations and GloVe for co-occurrence statistics, the ensemble model outperformed its individual components. This result is consistent with prior findings by Wu et al. (2022), Chen et al. (2024) and Nakhaee et al. (2024) in NLP applications in construction and reinforces the importance of multi-model strategies to handle the complexity and variability of construction language.

Furthermore, this work offers several key contributions to the field of construction cost estimation. First, it addresses a significant and underexplored challenge: the semantic alignment of QTO descriptions with standard cost references. Although prior research has examined NLP applications in cost management, ranging from predicting cost overruns from project descriptions (Williams and Gong 2014) to extracting reporting requirements from construction contracts to predict the associated time and costs (Jafari et al. 2021), very few studies have tackled the specific challenge of matching QTO items with standardized CCIs. Compared to Akanbi and Zhang (2021) who proposed a semantic matching model limited to selected work items, our study significantly expands the scope by testing the method across a broader set of classifications and demonstrating robust performance under real-world conditions. Also, compared to Tang et al. (2022), who focused on extracting structured information from textual work descriptions to support QTO automation through named entity recognition and rule-based methods, our study moves beyond information extraction to operationalize full semantic alignment with standardized cost databases. Specifically, our system works after QTO preparation and directly links QTO entries to relevant items in the cost libraries, enabling automated generation of BoQs with accurate cost assignments.

While NLP has been extensively studied in general-purpose applications, and Large Language Models (LLMs) have demonstrated strong capabilities in generic text understanding, this study

advances the field by tailoring and integrating multiple NLP methods to tackle a highly specific, domain-bound and operationally significant task in construction practice. The problem of aligning QTO text with structured cost entries in CCIs remains a practical challenge in quantity surveying, marked by inconsistencies in terminology, formatting and classification standards. The proposed system does not simply apply existing models out-of-the-box but adapts them to the construction context and combines their strengths through a structured, ensemble-based approach. In doing so, it demonstrates how domain-specific NLP configurations, when grounded in industry standards such as ASMM and AIQS BCI, can outperform more generic models in specialized tasks that require high accuracy and traceability. Importantly, the study presents a fully operational system that accepts QTO entries as input and generates a complete, itemized BoQ with associated costs in an accurate, efficient and user-friendly manner, supporting practical adoption in real-world workflows.

Our study operationalizes cost alignment as a practical, repeatable process that can be scaled and generalized. Compared to Deza et al. (2022), who applied supervised machine learning to classify UK infrastructure cost documents into ICMS categories for benchmarking, this study addresses everyday QS workflows in the Australian residential construction context. By working directly with ASMM and AIQS BCI, it enhances estimation workflows that rely on national standards and localized pricing data.

Overall, the high level of alignment accuracy of the developed ensemble model demonstrates the potential of NLP-based systems to significantly enhance the cost estimation process in the construction industry. Nevertheless, the developed AI-driven system is not intended to replace Qs but to function as a notification tool, alerting them to discrepancies between their judgments and the system's suggestions. This allows Qs to review and verify differences, whether due to system errors or human misjudgments, ultimately leading to more accurate cost estimates and minimizing potential financial losses. As emphasized by industry professionals, the integration of AI in construction must aim to augment existing processes rather than automate it entirely (Demchak 2022).

Integrating such AI-based approaches into the construction cost estimation process brings numerous advantages, including enhanced efficiency, reduced subjectivity and improved consistency in the provided estimations. Utilizing advanced NLP techniques not only enhances the accuracy and reliability of cost estimations but also streamlines the workflow, promoting a more systematic and data-driven approach to cost management in the AEC industry. This marks a significant step towards the broader adoption of AI technologies in construction management, facilitating more efficient and effective project management.

Model scalability and applicability across contexts

Although this study was tested on a single residential project using Melbourne-specific BCI cost data and ASMM classification, the developed NLP-based framework is designed to be adaptable and scalable across other geographic and classification contexts with minimal adjustment. Within Australia, the system can be directly applied to other capital cities by substituting the Melbourne-specific BCI benchmark field with those of other cities available in the AIQS BCI dataset. Since the model does not rely on location-specific training but rather on the semantic alignment between QTO descriptions and cost index entries, this substitution allows the model to remain fully operational without retraining.

Moreover, the framework can be extended to support alternative CCIs, such as the Rawlinsons Construction Cost Guide or CoreLogic's CCCI, with only minor preprocessing. As long as the structure of the input datasets, including QTO and CCI, remains consistent, the system can be executed with minimal reconfiguration. In cases where classification systems or formatting diverge, such as when using international standards like ICMS, UniFormat or MasterFormat, lightweight preprocessing steps (e.g. column renaming, unit normalization or mapping of classification hierarchies) may be required to synchronize the input data schemas.

Crucially, the ensemble approach used in this study is not trained exclusively on a fixed taxonomy or location but on the linguistic relationships between QTO entries and cost database descriptions. This allows the system to generalize across different datasets, provided that the QTO and CCI structures are aligned or made compatible through preprocessing. As such, while the validation was performed on a

single case study, the architecture and logic of the model are sufficiently flexible to support broader deployment across project types, building classifications, cost datasets and international standards.

Study limitations

In the case study, two classifications—Facade and Masonry—demonstrated lower agreement percentages. The lower accuracy for Masonry was attributed to the few work items in the QTO, which affected the model's performance in assigning relevant work descriptions. This scarcity of data points can make it challenging for NLP models to accurately capture the full scope of work item descriptions, leading to less reliable cost matches. Essentially, fewer data points mean less context for the model to accurately interpret and match descriptions, which can hinder its performance. Conversely, a large number of work items, totaling 153, in the Facade classification presented a different challenge. The vast array of items led to a high level of diversity and complexity within this classification, making it more difficult for the NLP models to find precise matches. This complexity can introduce a higher variance in the text descriptions, increasing the difficulty of consistently identifying and aligning with the correct cost entries.

Moreover, the study focused on a selected subset of classifications within the ASMM that had sufficient representation in the dataset and corresponding entries in the AIQS BCI. Classifications with limited quantified work items or missing BCI cost references were excluded to avoid introducing sparsity and noise that could compromise model evaluation. The ensemble model achieved consistently high accuracy across the included classifications, demonstrating the viability of NLP-based methods for aligning QTO descriptions with industry-standard cost items. The results provide compelling evidence that semantic alignment using AI-driven techniques can significantly reduce manual workload and enhance the reliability of cost estimation processes. These outcomes validate the core methodological contribution of this paper and establish a strong foundation for broader applications. Nonetheless, it is acknowledged that this exclusion, while methodologically necessary, could raise concerns about the reliability and generalizability of the model across the full spectrum of cost classifications encountered in practice. While the ensemble model showed promising results within this scope, its extension to other classification systems or CCIs may require further validation and potential methodology adaptation.

Additionally, although extensive preprocessing steps were undertaken, such as text normalization, stop word removal and handling plurals, there remains a risk of losing critical context or nuances that could affect the accuracy of the matches. The preprocessing steps may not fully capture the intricacies of construction terminology and descriptions, impacting the final alignment accuracy. Additionally, the case study focused on a specific project involving a 13-level residential building. The findings and model performance might not fully represent the complexities and variability encountered in other types of construction projects or different scales of operations. Therefore, while the results are promising, the applicability of the proposed system to a broader range of projects remains to be tested.

Future work

In order to address the above-mentioned limitations, future research could focus on expanding the case study to other construction projects, encompassing different building types, materials and workflows. Including more diverse and extensive classifications and work items, as well as other CCIs like the Rawlinsons Construction Cost Guide and CoreLogic's CCCI, which include more detailed work items and cost descriptions, can improve the models' ability to generalize and accurately capture the nuances of construction descriptions. This expansion could lead to the development of domain-specific language models tailored to the construction industry.

Besides, incorporating feedback from a set of QS professionals through mixed-approach methodologies could be useful for refining the models. This feedback can help bridge the gap between automated systems and human expertise, ensuring more accurate and reliable cost estimations. Engaging with QS professionals can also provide valuable insights into the practical challenges and nuances of cost estimation, further enhancing the system's applicability and accuracy.

Future research can also explore the international application of the developed system by extending the methodology to integrate with other classification and measurement standards and cost datasets, thereby broadening the system's applicability and robustness. Developing hybrid models that combine NLP-based methods with deep learning approaches can also enhance the models' capability to handle complex and diverse construction data. By incorporating advanced techniques and broader datasets, the system can be made more robust and adaptable to various contexts in the construction industry.

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Data availability statement

The authors do not have permission to share data.

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